

Nathanya Xavier Napitupulu

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SUMMARY

Computer Science Undergraduate at Bina Nusantara University with hands-on experience in frontend development, software engineering, UI/UX design, and graphic design through academic projects and professional experience. Experienced in developing web-based applications using Python, Java, HTML, CSS, SQL and JavaScript. Passionate about artificial intelligence and web development, with a strong commitment to continuous learning, collaboration, and building user-centered solutions.

EDUCATION

BINUS University | GPA: 3.10/4.00

Bachelor of Science in Computer Science, Specialization in Intelligence Systems

Jakarta, Indonesia

Aug 2024 – Present

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C, SQL.

Web Development: HTML, CSS, Django, Streamlit, API Design.

Machine Learning & Deep Learning: YOLOv8, PyTorch, CNNs, Object Detection, Transfer Learning, Computer Vision.

Framework & Tools: NumPy, Pandas, Matplotlib, OpenCV, Git, GitHub, SQLite.

Graphic Design & Illustration: Figma, Canva, IbisPaint.

Soft Skills: Cross-functional Leadership, Coaching & Mentoring, Teamwork, Stakeholder Management, Collaboration & Communication.

SELECTED PROJECTS

AgriYield - Rice Yield Prediction

Feb 2026 – June 2026

Machine Learning Developer & Front-End Developer

Python, PowerShell, Streamlit, C++, C, Git, GitHub

- Developed a machine learning-based rice yield prediction system using 2,338 agricultural samples and 45 original features, including preprocessing, feature selection, model training, and evaluation.
- Compared Linear Regression, KNN, and Random Forest models, with the optimized Random Forest achieving the best performance of $R^2 = 0.9903$, MAE = 657.62 kg, and RMSE = 912.78 kg, with a Train-Test R^2 gap of 0.0019.
- Developed and integrated a Streamlit-based web application to provide interactive and real-time rice yield predictions.

Schola - Scholarship Management System

Feb 2026 – June 2026

Software Engineering Developer

HTML, CSS, Python, JavaScript

- Developed a web-based scholarship management platform using Python, HTML, CSS, and JavaScript, enabling students to browse scholarships, submit applications, upload required documents, and track their application status.
- Developed separate functionalities for students and administrators, allowing administrators to manage scholarship information, review applications, and monitor the overall application process.

Comparative Analysis of Pothole Detection and Dimensional Estimation

Feb 2026 – June 2026

Computer Vision Research & AI Developer

Jupyter Notebook, Python, Java, Swift, C++, CMake

- Developed and compared Traditional Computer Vision and YOLOv8-Seg approaches for pothole detection and segmentation, achieving a mean IoU of 0.1521 and 0.4831 respectively.
- Evaluated segmentation performance using IoU and Dice Score, with YOLOv8-Seg achieving a maximum IoU of 0.9528 and maximum Dice Score of 0.9758, outperforming the Traditional CV approach.
- Integrated MiDaS depth estimation with YOLOv8-Seg to estimate pothole dimensions, producing an estimated width of 87.23 cm compared with 26.0 cm using manual Y-axis estimation.

Floral and Co - E Commerce Website for Plant Shopping

Feb 2025 – June 2025

Human and Computer Interaction Developer

Figma, HTML, CSS, JavaScript.

- Conducted user requirement analysis and designed the complete user experience, including wireframing, interface design, and responsive layouts based on Human Computer Interaction (HCI) principles.
- Developed the front-end implementation of the e-commerce website, focusing on usability, accessibility, visual consistency, and intuitive navigation to support efficient product browsing and shopping.

ParkEasy - Smart Parking Mobile Application

Sept 2024 – Dec 2024

Program Design Methods Developer

Figma, UML, Use Case Diagram, UI/UX Design.

- Analyzed user requirements and designed the system workflow using UML Use Case Diagrams and user flows for a smart parking application.
- Designed an interactive mobile application prototype in Figma, supporting parking reservation, arrival and departure scheduling, voucher management, payment options, and parking time tracking.

Nathanya Website

June 2026 – Present

Personal UI/UX Designer & Front-End Developer

HTML, CSS, Git, GitHub.

- Designed and developed a highly interactive personal portfolio website inspired by modern desktop operating systems and window management interface, focusing on responsive design and user experience.
- Implemented dynamic window style interactions and overlay modals for sections such as About, CV, Links, Projects, and Contact, including light and dark mode functionality.

EXPERIENCE

Head of Registration Management

Sept 2023 – Sept 2023

Digital Literacy Event - "Netizen Siap Antisipasi Hoax Pemilu 2024"

Jakarta, Indonesia

- Coordinated participant registration and managed administrative processes for a national digital literacy event organized by the Ministry of Communication and Information Technology (Kominfo RI).
- Managed registration data for 200+ participants and coordinated administrative processes to support smooth event operations.

Freelance Graphic Designer

March 2024 – Present

Maybaker & Mami Mina - "The Salted Sea" Food Brand

Jakarta, Indonesia

- Designed branding materials and social media content using Figma to support the brand's visual identity.
- Collaborated with clients to maintain brand consistency across promotional and digital content.

International Sales Promotion Representative

March 2024 – Present

Jakarta International Expo - "Korean DOBU Mask"

Jakarta, Indonesia

- Represented the DOBU respiratory protection brand at an international exhibition by introducing product and explaining their features and proper usage.
- Assisted international customers in product selection while providing professional customer service and sales support.

ORGANIZATION & LEADERSHIP

Purna Paskibraka & Training Mentor

Aug 2022 – Aug 2023

South Jakarta Mayor Level Paskibraka

Jakarta, Indonesia

- Selected as a member of the South Jakarta Mayor Level Paskibraka for the 77th Independence Day Ceremony.
- Served as a training mentor for junior members and participated in FOPASBIN at BINUS University for the 80th Independence Day Ceremony.

Public Relations & Graphic Designer - Leading Member

Nov 2022 – Aug 2023

SPLASH (Sama Paskibraka Lomba Seharian)

Jakarta, Indonesia

- Served as a leading member in the Public Relations and Graphic Design division.
- Created promotional materials and collaborated on event branding and publication content.

Social Activity Fun Learning - PAUD & HIMTI Organization

June 2025 – Feb 2026

BINUS Community Service (COMSERV)

Jakarta, Indonesia

- Participated in a community service program by delivering interactive learning activities for early childhood students.
- Created graphic design materials to support HIMTI organizational events.

CERTIFICATIONS

Fluent in Java and JavaScript - UdeMy <i>Completed an 11 hour online course covering the fundamentals of both programming.</i>	Jakarta, Indonesia March 2021 – June 2021
Binus International Week 2025 - BINUS University <i>Participated in the Master Studies in Germany & Scholarship Opportunities session presented by DAAD and BINUS Internationalization.</i>	Jakarta, Indonesia Feb 2025
CS50's Introduction to Artificial Intelligence with Python - Harvard University <i>Completed Harvard's CS50 course focused on the fundamentals and applications of artificial intelligence using Python.</i>	Jakarta, Indonesia Jan 2023
Head of Registration Team Certificate - Kominfo <i>Contributed as part of the Registration Team for the Netizen Desa Siap Antisipasi Hoax Pemilu 2024 digital literacy event in Jakarta.</i>	Jakarta, Indonesia Nov 2023
Complete Python Programming Masterclass: Beginner to Advanced - UdeMy <i>Completed a Python programming course covering concepts from beginner to advanced levels.</i>	Jakarta, Indonesia March 2022
Certificate of Appreciation HIMTI - BINUS / TECHNO <i>Received a Certificate of Appreciation as a member of DnD Division of contributing to the HIMTI with TECHNO event.</i>	Jakarta, Indonesia Sept 2025
ClimateScience Crash Course <i>Completed the ClimateScience Crash Course, covering fundamental knowledge and concepts related to climate science.</i>	Jakarta, Indonesia March 2023
TECHFEST 2025 Certificate <i>Received a certificate for participation in TECHFEST 2025: "MAGIC: Mastering Analytics & Graphics in Innovation Competitions" held by HIMTI Binus University.</i>	Jakarta, Indonesia Sept 2025
Certificate of Appreciation PASKIBRAKA - South Jakarta Mayor's Office <i>Received a Certificate as a PASKIBRAKA member of the 77th Independency Day Ceremony of the Republic of Indonesia at the South Jakarta Mayor Office.</i>	Jakarta, Indonesia Aug 2022
International Essay Contest for Young People in 2021 & 2022 <i>Received a Certificate of Participation from The Goi Peace Foundation for submitting an essay to the 2021 Aug 2021 & Aug 2022 & 2022 international Essay Contest for Young People.</i>	Jakarta, Indonesia Aug 2021 & Aug 2022
National Physics Olympiad - Final <i>Participated as a finalist in the National Physics Olympiad</i>	Jakarta, Indonesia Feb 2022

LANGUAGES

English (Professional Working Proficiency)	German (Beginner)
Indonesian (Native)	Chinese (Beginner)